# SQL Project: Online Book Store

*Practical Case Studies & SQL Solutions*

## Basic Queries

1. Retrieve all books in the "Fiction" genre
2. Find books published after the year 1950
3. List all customers from Canada
4. Show orders placed in November 2023
5. Retrieve the total stock of books available
6. Find the details of the most expensive book
7. Show all customers who ordered more than 1 quantity of a book
8. Retrieve all orders where the total amount exceeds $20
9. List all genres available in the Books table
10. Find the book with the lowest stock
11. Calculate the total revenue generated from all orders

## Advanced Queries

1. Retrieve the total number of books sold for each genre
2. Find the average price of books in the "Fantasy" genre
3. List customers who have placed at least 2 orders
4. Find the most frequently ordered book
5. Show the top 3 most expensive books of 'Fantasy' Genre
6. Retrieve the total quantity of books sold by each author
7. List the cities where customers who spent over $30 are located
8. Find the customer who spent the most on orders
9. Calculate the stock remaining after fulfilling all orders

by Noor Mohammad

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**DATABASE SETUP**

**Task to do:** Create Customers and Orders tables with primary and foreign keys

Query | Query History

```
    Email VARCHAR(100),
    Phone VARCHAR(15),
    City VARCHAR(50),
    Country VARCHAR(150)
);
DROP TABLE IF EXISTS orders;
CREATE TABLE Orders (
    Order_ID SERIAL PRIMARY KEY,
    Customer_ID INT REFERENCES Customers(Customer_ID),
    Book_ID INT REFERENCES Books(Book_ID),
    Order_Date DATE,
    Quantity INT,
    Total_Amount NUMERIC(10, 2)
);

SELECT * FROM Books;
SELECT * FROM Customers;
SELECT * FROM Orders;
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Configurable modular throughput | Joseph Crane | Biography | 1949 | 21.34 | 100 |
| 2 | 2 | Persevering reciprocal knowledge user | Mario Moore | Fantasy | 1971 | 35.80 | 19 |
| 3 | 3 | Streamlined coherent initiative | Derrick Howard | Non-Fiction | 1913 | 15.75 | 27 |
| 4 | 4 | Customizable 24hour product | Christopher Andrews | Fiction | 2020 | 43.52 | 8 |
| 5 | 5 | Adaptive 5thgeneration encoding | Juan Miller | Fantasy | 1956 | 10.95 | 16 |

Total rows: 500 | Query complete 00:00:00.252 | CRLF | Ln 31, Col 1

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**DATA VERIFICATION**

**Task to do:** Verify data ingestion for the Customers table

Query | Query History

```
    Email VARCHAR(100),
    Phone VARCHAR(15),
    City VARCHAR(50),
    Country VARCHAR(150)
);
DROP TABLE IF EXISTS orders;
CREATE TABLE Orders (
    Order_ID SERIAL PRIMARY KEY,
    Customer_ID INT REFERENCES Customers(Customer_ID),
    Book_ID INT REFERENCES Books(Book_ID),
    Order_Date DATE,
    Quantity INT,
    Total_Amount NUMERIC(10, 2)
);

SELECT * FROM Books;
SELECT * FROM Customers;
SELECT * FROM Orders;
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | customer_id [PK] integer | name character varying (100) | email character varying (100) | phone character varying (15) | city character varying (50) | country character varying (150) |
|---|---|---|---|---|---|---|
| 1 | 1 | Deborah Griffith | balljoseph@wright-keith.net | 1234567891 | South Craigfort | Denmark |
| 2 | 2 | Crystal Clements | kimberlybennett@curtis.com | 1234567892 | East Derekberg | Nicaragua |
| 3 | 3 | Susan Fuller | beanmichael@burnett-stewart.net | 1234567893 | Austinbury | Equatorial Guinea |
| 4 | 4 | Jamie Ramirez | amandahood@warren.com | 1234567894 | Dianamouth | Slovenia |
| 5 | 5 | Marcus Murphy | connerjohn@yahoo.com | 1234567895 | Smithbury | Guinea-Bissau |

Total rows: 500 | Query complete 00:00:00.180 | CRLF | Ln 32, Col 1

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**DATA VERIFICATION**

**Task to do:** Verify data ingestion for the Orders table

Query | Query History

```
16      Email VARCHAR(100),
17      Phone VARCHAR(15),
18      City VARCHAR(50),
19      Country VARCHAR(150)
20  );
21  DROP TABLE IF EXISTS orders;
22  CREATE TABLE Orders (
23      Order_ID SERIAL PRIMARY KEY,
24      Customer_ID INT REFERENCES Customers(Customer_ID),
25      Book_ID INT REFERENCES Books(Book_ID),
26      Order_Date DATE,
27      Quantity INT,
28      Total_Amount NUMERIC(10, 2)
29  );
30
31  SELECT * FROM Books;
32  SELECT * FROM Customers;
33  SELECT * FROM Orders;
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | order_id [PK] integer | customer_id integer | book_id integer | order_date date | quantity integer | total_amount numeric (10,2) |
|---|---|---|---|---|---|---|
| 1 | 1 | 84 | 169 | 2023-05-26 | 8 | 188.56 |
| 2 | 2 | 137 | 301 | 2023-01-23 | 10 | 216.60 |
| 3 | 3 | 216 | 261 | 2024-05-27 | 6 | 85.50 |
| 4 | 4 | 433 | 343 | 2023-11-25 | 7 | 301.21 |
| 5 | 5 | 14 | 431 | 2023-07-26 | 7 | 136.36 |

Total rows: 500 | Query complete 00:00:00.185 | CRLF | Ln 33, Col 1

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**DATA IMPORT**

**Task to do:** Bulk import Books, Customers, and Orders datasets from CSV files

Query | Query History

```
31  SELECT * FROM Books;
32  SELECT * FROM Customers;
33  SELECT * FROM Orders;
34
35
36  -- Import Data into Books Table
37  COPY Books(Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
38  FROM 'C:\Users\Public\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Books.csv'
39  CSV HEADER;
40
41  -- Import Data into Customers Table
42  COPY Customers(Customer_ID, Name, Email, Phone, City, Country)
43  FROM 'C:\Users\Public\Downloads\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Customers.csv'
44  CSV HEADER;
45
46  -- Import Data into Orders Table
47  COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
48  FROM 'C:\Users\Public\Downloads\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Orders.csv'
49  CSV HEADER;
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Configurable modular throughput | Joseph Crane | Biography | 1949 | 21.34 | 100 |
| 2 | 2 | Persevering reciprocal knowledge user | Mario Moore | Fantasy | 1971 | 35.80 | 19 |
| 3 | 3 | Streamlined coherent initiative | Derrick Howard | Non-Fiction | 1913 | 15.75 | 27 |
| 4 | 4 | Customizable 24hour product | Christopher Andrews | Fiction | 2020 | 43.52 | 8 |
| 5 | 5 | Adaptive 5thgeneration encoding | Juan Miller | Fantasy | 1956 | 10.95 | 16 |

Total rows: 500 | Query complete 00:00:00.165 | CRLF | Ln 31, Col 1

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**BASIC QUERY 1**

**Task to do:** Retrieve all books in the 'Fiction' genre

Query | Query History

```
38  FROM 'C:\Users\Public\ST - SQL ALL PRACTICE FILES-2\All Ex
39  CSV HEADER;
40
41  -- Import Data into Customers Table
42  COPY Customers(Customer_ID, Name, Email, Phone, City, Country)
43  FROM 'C:\Users\Public\Downloads\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Customers.csv'
44  CSV HEADER;
45
46  -- Import Data into Orders Table
47  COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
48  FROM 'C:\Users\Public\Downloads\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Orders.csv'
49  CSV HEADER;
50
51
52  -- 1) Retrieve all books in the "Fiction" genre:
53
54  SELECT * FROM Books
55  WHERE Genre='Fiction';
56
```

Data Output | Messages | Notifications

Showing rows: 1 to 60 | Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 4 | Customizable 24hour product | Christopher Andrews | Fiction | 2020 | 43.52 | 8 |
| 2 | 22 | Multi-layered optimizing migration | Wesley Escobar | Fiction | 1908 | 39.23 | 78 |
| 3 | 28 | Expanded analyzing portal | Lisa Coffey | Fiction | 1941 | 37.51 | 79 |
| 4 | 29 | Quality-focused multi-tasking challenge | Katrina Underwood | Fiction | 1905 | 31.12 | 100 |
| 5 | 31 | Implemented encompassing conglomerat... | Melissa Taylor | Fiction | 2010 | 21.23 | 44 |

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**BASIC QUERY 2**

**Task to do:** Find books published after the year 1950

Query | Query History

```
46  -- Import Data into Orders Table
47  COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
48  FROM 'C:\Users\Public\Downloads\ST - SQL ALL PRACTICE FILES-2\All Excel Practice Files\Orders.csv'
49  CSV HEADER;
50
51
52  -- 1) Retrieve all books in the "Fiction" genre:
53
54  SELECT * FROM Books
55  WHERE Genre='Fiction';
56
57  -- 2) Find books published after the year 1950:
58  SELECT * FROM Books
59  WHERE Published_year>1950;
60
```

Data Output | Messages | Notifications

Showing rows: 1 to 292 Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 2 | Persevering reciprocal knowledge user | Mario Moore | Fantasy | 1971 | 35.80 | 19 |
| 2 | 4 | Customizable 24hour product | Christopher Andrews | Fiction | 2020 | 43.52 | 8 |
| 3 | 5 | Adaptive 5thgeneration encoding | Juan Miller | Fantasy | 1956 | 10.95 | 16 |
| 4 | 6 | Advanced encompassing implementation | Bryan Morgan | Biography | 1985 | 6.56 | 2 |
| 5 | 8 | Persistent local encoding | Troy Cox | Science Fiction | 2019 | 48.99 | 84 |
| 6 | 9 | Optimized interactive challenge | Colin Buckley | Fantasy | 1987 | 14.33 | 70 |
| 7 | 10 | Ergonomic national hub | Samantha Ruiz | Mystery | 2015 | 24.63 | 25 |
| 8 | 11 | Secured zero tolerance time-frame | Denise Barnes | Fantasy | 1998 | 35.95 | 10 |

Total rows: 292 Query complete 00:00:00.233 CRLF Ln 59, Col 27

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | SQL Project OnlineBookstore.sql*

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No limit

**BASIC QUERY 3**

**Task to do:** List all customers from Canada

Query | Query History

```
50
51
52  -- 1) Retrieve all books in the "Fiction" genre:
53
54  SELECT * FROM Books
55  WHERE Genre='Fiction';
56
57  -- 2) Find books published after the year 1950:
58  SELECT * FROM Books
59  WHERE Published_year>1950;
60
61  -- 3) List all customers from the Canada:
62  SELECT * FROM Customers
63  WHERE country='Canada';
64
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

| | customer_id [PK] integer | name character varying (100) | email character varying (100) | phone character varying (15) | city character varying (50) | country character varying (150) |
|---|---|---|---|---|---|---|
| 1 | 38 | Nicholas Harris | christine93@perkins.com | 1234567928 | Davistown | Canada |
| 2 | 415 | James Ramirez | robert54@hall.com | 1234568305 | Maxwelltown | Canada |
| 3 | 468 | David Hart | stokesrebecca@gmail.c… | 1234568358 | Thompsonfurt | Canada |

Total rows: 3 | Query complete 00:00:00.107 | CRLF | Ln 63, Col 24

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**BASIC QUERY 4**

**Task to do:** Show orders placed in November 2023

Query | Query History

```
54  SELECT * FROM Books
55  WHERE Genre='Fiction';
56
57  -- 2) Find books published after the year 1950:
58  SELECT * FROM Books
59  WHERE Published_year>1950;
60
61  -- 3) List all customers from the Canada:
62  SELECT * FROM Customers
63  WHERE country='Canada';
64
65  -- 4) Show orders placed in November 2023:
66
67  SELECT * FROM Orders
68  WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';
```

Data Output | Messages | Notifications

Showing rows: 1 to 25 | Page No: 1 of 1

| | order_id [PK] integer | customer_id integer | book_id integer | order_date date | quantity integer | total_amount numeric (10,2) |
|---|---|---|---|---|---|---|
| 1 | 4 | 433 | 343 | 2023-11-25 | 7 | 301.21 |
| 2 | 19 | 496 | 60 | 2023-11-17 | 9 | 316.26 |
| 3 | 75 | 291 | 375 | 2023-11-30 | 5 | 170.75 |
| 4 | 132 | 469 | 333 | 2023-11-22 | 7 | 194.32 |
| 5 | 137 | 474 | 471 | 2023-11-25 | 8 | 363.04 |
| 6 | 163 | 207 | 384 | 2023-11-23 | 3 | 101.76 |
| 7 | 182 | 129 | 293 | 2023-11-01 | 7 | 125.51 |
| 8 | 200 | 313 | 303 | 2023-11-23 | 1 | 6.57 |

Successfully run. Total query runtime: 392 msec. 25 rows affected.

Total rows: 25 | Query complete 00:00:00.392 | CRLF | Ln 68, Col 56

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**BASIC QUERY 5**

**Task to do:** Retrieve the total stock of books available

Query | Query History

```
60
61  -- 3) List all customers from the Canada:
62  SELECT * FROM Customers
63  WHERE country='Canada';
64
65  -- 4) Show orders placed in November 2023:
66
67  SELECT * FROM Orders
68  WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';
69
70  -- 5) Retrieve the total stock of books available:
71
72  SELECT SUM(stock) AS Total_Stock
73  From Books;
74
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | total_stock bigint |
|---|---|
| 1 | 25056 |

Total rows: 1 | Query complete 00:00:00.166 | CRLF | Ln 73, Col 12

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | SQL Project OnlineBookstore.sql*

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**BASIC QUERY 6**

**Task to do:** Find the details of the most expensive book

Query | Query History

```
67  SELECT * FROM Orders
68  WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';
69
70  -- 5) Retrieve the total stock of books available:
71
72  SELECT SUM(stock) AS Total_Stock
73  From Books;
74
75
76  -- 6) Find the details of the most expensive book:
77  SELECT * FROM Books
78  ORDER BY Price DESC;
79  LIMIT 1; (if i want only one highest)
80
81
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 340 | Proactive system-worthy orchestration | Robert Scott | Mystery | 1907 | 49.98 | 88 |
| 2 | 155 | Optimized content-based standardization | Timothy Adams | Science Fiction | 1901 | 49.96 | 88 |
| 3 | 240 | Stand-alone content-based hub | Lisa Ellis | Fantasy | 1957 | 49.90 | 41 |
| 4 | 100 | Synchronized client-server service-desk | James Alvarado | Fiction | 1906 | 49.89 | 29 |
| 5 | 119 | Switchable modular moratorium | Tonya Saunders | Romance | 2010 | 49.88 | 76 |
| 6 | 260 | Business-focused methodical application | Brian King | Fiction | 1907 | 49.59 | 10 |
| 7 | 95 | Function-based heuristic analyzer | Craig Smith | Biography | 1990 | 49.53 | 11 |
| 8 | 49 | Robust attitude-oriented attitude | Zachary Hayes | Biography | 1955 | 49.50 | 15 |

Total rows: 500 | Query complete 00:00:00.170 | CRLF | Ln 78, Col 21

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**BASIC QUERY 7**

**Task to do:** Show all customers who ordered more than 1 quantity of a book

Query | Query History

```
70  -- 5) Retrieve the total stock of books available:
71
72  SELECT SUM(stock) AS Total_Stock
73  From Books;
74
75
76  -- 6) Find the details of the most expensive book:
77  SELECT * FROM Books
78  ORDER BY Price DESC;
79  LIMIT 1; (if i want only one highest)
80
81
82  -- 7) Show all customers who ordered more than 1 quantity of a book:
83  SELECT * FROM Orders
84  WHERE quantity>1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 438 | Page No: 1 of 1

| | order_id [PK] integer | customer_id integer | book_id integer | order_date date | quantity integer | total_amount numeric (10,2) |
|---|---|---|---|---|---|---|
| 1 | 1 | 84 | 169 | 2023-05-26 | 8 | 188.56 |
| 2 | 2 | 137 | 301 | 2023-01-23 | 10 | 216.60 |
| 3 | 3 | 216 | 261 | 2024-05-27 | 6 | 85.50 |
| 4 | 4 | 433 | 343 | 2023-11-25 | 7 | 301.21 |
| 5 | 5 | 14 | 431 | 2023-07-26 | 7 | 136.36 |
| 6 | 6 | 439 | 119 | 2024-10-11 | 5 | 249.40 |
| 7 | 7 | 195 | 467 | 2023-10-23 | 6 | 82.92 |
| 8 | 8 | 32 | 159 | 2024-05-07 | 4 | 144.84 |

Total rows: 438 | Query complete 00:00:00.201 | CRLF | Ln 84, Col 18

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No limit

**BASIC QUERY 8**

**Task to do:** Retrieve all orders where the total amount exceeds $20

Query | Query History

```
79  LIMIT 1; (if i want only one highest)
80
81
82  -- 7) Show all customers who ordered more than 1 quantity of a book:
83  SELECT * FROM Orders
84  WHERE quantity>1;
85
86
87
88  -- 8) Retrieve all orders where the total amount exceeds $20:
89  SELECT * FROM Orders
90  WHERE total_amount>20;
91
92
93
```

Data Output | Messages | Notifications

Showing rows: 1 to 473 | Page No: 1 of 1

| | order_id [PK] integer | customer_id integer | book_id integer | order_date date | quantity integer | total_amount numeric (10,2) |
|---|---|---|---|---|---|---|
| 1 | 1 | 84 | 169 | 2023-05-26 | 8 | 188.56 |
| 2 | 2 | 137 | 301 | 2023-01-23 | 10 | 216.60 |
| 3 | 3 | 216 | 261 | 2024-05-27 | 6 | 85.50 |
| 4 | 4 | 433 | 343 | 2023-11-25 | 7 | 301.21 |
| 5 | 5 | 14 | 431 | 2023-07-26 | 7 | 136.36 |
| 6 | 6 | 439 | 119 | 2024-10-11 | 5 | 249.40 |
| 7 | 7 | 195 | 467 | 2023-10-23 | 6 | 82.92 |
| 8 | 8 | 32 | 159 | 2024-05-07 | 4 | 144.84 |

Total rows: 473 | Query complete 00:00:01.172 | CRLF | Ln 90, Col 23

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | SQL Project OnlineBookstore.sql*

OnlineBookstore SQL Project/postgres@PostgreSQL 18

No limit

**BASIC QUERY 9**

**Task to do:** List all genres available in the Books table

Query | Query History

```
83  SELECT * FROM Orders
84  WHERE quantity>1;
85
86
87
88  -- 8) Retrieve all orders where the total amount exceeds $20:
89  SELECT * FROM Orders
90  WHERE total_amount>20;
91
92
93
94  -- 9) List all genres available in the Books table:
95  SELECT DISTINCT genre FROM Books;
96
97
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 | Page No: 1 of 1

| | genre character varying (50) |
|---|---|
| 1 | Romance |
| 2 | Biography |
| 3 | Mystery |
| 4 | Fantasy |
| 5 | Fiction |
| 6 | Non-Fiction |
| 7 | Science Fiction |

Total rows: 7 | Query complete 00:00:00.182 | CRLF | Ln 95, Col 34

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OnlineBookstore SQL Project/postgres@PostgreSQL 18

No limit

BASIC QUERY 10

**Task to do:** Find the book with the lowest stock

Query | Query History

```
-- 8) Retrieve all orders where the total amount exceeds $.
SELECT * FROM Orders
WHERE total_amount>20;



-- 9) List all genres available in the Books table:
SELECT DISTINCT genre FROM Books;


-- 10) Find the book with the lowest stock:
SELECT * FROM Books
ORDER BY stock
LIMIT 1;

```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 44 | Networked systemic implementati... | Ryan Frank | Science Fiction | 1965 | 13.55 | 0 |

Total rows: 1 | Query complete 00:00:00.191 | CRLF | Ln 101, Col 9

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | SQL Project OnlineBookstore.sql*

OnlineBookstore SQL Project/postgres@PostgreSQL 18

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**BASIC QUERY 11**

**Task to do:** Calculate the total revenue generated from all orders

Query | Query History

```
93
94  -- 9) List all genres available in the Books table:
95  SELECT DISTINCT genre FROM Books;
96
97
98  -- 10) Find the book with the lowest stock:
99  SELECT * FROM Books
100 ORDER BY stock
101 LIMIT 1;
102
103
104 -- 11) Calculate the total revenue generated from all orders:
105 SELECT SUM(total_amount) As Revenue
106 FROM Orders;
107
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | revenue numeric |
|---|---|
| 1 | 75628.66 |

Total rows: 1 | Query complete 00:00:00.174 | CRLF | Ln 107, Col 1

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | SQL Project OnlineBookstore.sql*

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**ADVANCED QUERY 1**

**Task to do:** Retrieve the total number of books sold for each genre

Query | Query History

```
105 SELECT SUM(total_amount) As Revenue
106 FROM Orders;
107
108 -- Advance Questions :
109
110 -- 1) Retrieve the total number of books sold for each genre:
111
112 SELECT * FROM ORDERS;
113
114 SELECT b.Genre, SUM(o.Quantity) AS Total_Books_sold
115 FROM Orders o
116 JOIN Books b ON o.book_id = b.book_id
117 GROUP BY b.Genre;
118
119
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 Page No: 1 of 1

| | genre character varying (50) | total_books_sold bigint |
|---|---|---|
| 1 | Romance | 439 |
| 2 | Biography | 285 |
| 3 | Mystery | 504 |
| 4 | Fantasy | 446 |
| 5 | Fiction | 225 |
| 6 | Non-Fiction | 351 |
| 7 | Science Fiction | 447 |

Total rows: 7 | Query complete 00:00:00.104 | CRLF | Ln 117, Col 18

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No limit

ADVANCED QUERY 2

**Task to do:** Find the average price of books in the 'Fantasy' genre

Query | Query History

```
110 -- 1) Retrieve the total number of books sold for each ge
111
112 SELECT * FROM ORDERS;
113
114 SELECT b.Genre, SUM(o.Quantity) AS Total_Books_sold
115 FROM Orders o
116 JOIN Books b ON o.book_id = b.book_id
117 GROUP BY b.Genre;
118
119
120 -- 2) Find the average price of books in the "Fantasy" genre:
121 SELECT AVG(price) AS Average_Price
122 FROM Books
123 WHERE Genre = 'Fantasy';
124
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | average_price numeric |
|---|---|
| 1 | 25.9816901408450704 |

Successfully run. Total query runtime: 122 msec. 1 rows affected.

Total rows: 1 | Query complete 00:00:00.122 | CRLF | Ln 123, Col 25

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ADVANCED QUERY 3

**Task to do:** List customers who have placed at least 2 orders

Query | Query History

```
119
120 -- 2) Find the average price of books in the "Fantasy" genre:
121 SELECT AVG(price) AS Average_Price
122 FROM Books
123 WHERE Genre = 'Fantasy';
124
125
126 -- 3) List customers who have placed at least 2 orders:
127 SELECT o.customer_id, c.name, COUNT(o.Order_id) AS ORDER_COUNT
128 FROM orders o
129 JOIN customers c ON o.customer_id=c.customer_id
130 GROUP BY o.customer_id, c.name
131 HAVING COUNT(Order_id) >=2;
132
133
```

Data Output | Messages | Notifications

Showing rows: 1 to 139 Page No: 1 of 1

| | customer_id integer | name character varying (100) | order_count bigint |
|---|---|---|---|
| 1 | 225 | Christopher Mccullough | 2 |
| 2 | 418 | Kiara Blankenship MD | 3 |
| 3 | 322 | William Cameron | 3 |
| 4 | 325 | Emily Vargas | 4 |
| 5 | 376 | Justin Donaldson | 2 |
| 6 | 486 | Melanie Kelly | 2 |
| 7 | 461 | Crystal Pierce | 3 |
| 8 | 2 | Crystal Clements | 2 |

Total rows: 139 Query complete 00:00:00.094 CRLF Ln 132, Col 1

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ADVANCED QUERY 4

**Task to do:** Find the most frequently ordered book

Query | Query History

```
128 FROM orders o
129 JOIN customers c ON o.customer_id=c.customer_id
130 GROUP BY o.customer_id, c.name
131 HAVING COUNT(Order_id) >=2;
132
133
134
135
136 -- 4) Find the most frequently ordered book:
137 SELECT o.Book_id, b.title, COUNT(o.order_id) AS ORDER_COUNT
138 FROM orders o
139 JOIN books b ON o.book_id=b.book_id
140 GROUP BY o.book_id, b.title
141 ORDER BY ORDER_COUNT DESC LIMIT 1;
142
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | book_id integer | title character varying (100) | order_count bigint |
|---|---|---|---|
| 1 | 88 | Robust tangible hardware | 4 |

Total rows: 1 | Query complete 00:00:00.151 | CRLF | Ln 141, Col 35

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**ADVANCED QUERY 5**

**Task to do:** Show the top 3 most expensive books of 'Fantasy' Genre

Query | Query History

```
136  -- 4) Find the most frequently ordered book:
137  SELECT o.Book_id, b.title, COUNT(o.order_id) AS ORDER_COUNT
138  FROM orders o
139  JOIN books b ON o.book_id=b.book_id
140  GROUP BY o.book_id, b.title
141  ORDER BY ORDER_COUNT DESC LIMIT 1;
142
143
144
145  -- 5) Show the top 3 most expensive books of 'Fantasy' Genre :
146  SELECT * FROM books
147  WHERE genre ='Fantasy'
148  ORDER BY price DESC LIMIT 3;
149
150
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | author character varying (100) | genre character varying (50) | published_year integer | price numeric (10,2) | stock integer |
|---|---|---|---|---|---|---|---|
| 1 | 240 | Stand-alone content-based hub | Lisa Ellis | Fantasy | 1957 | 49.90 | 41 |
| 2 | 462 | Innovative 3rdgeneration database | Allison Contreras | Fantasy | 1988 | 49.23 | 62 |
| 3 | 238 | Optimized even-keeled analyzer | Sherri Griffith | Fantasy | 1975 | 48.97 | 72 |

Total rows: 3 | Query complete 00:00:00.175 | CRLF | Ln 148, Col 29

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ADVANCED QUERY 6

**Task to do:** Retrieve the total quantity of books sold by each author

Query | Query History

```
144
145  -- 5) Show the top 3 most expensive books of 'Fantasy' Genre :
146  SELECT * FROM books
147  WHERE genre ='Fantasy'
148  ORDER BY price DESC LIMIT 3;
149
150
151  -- 6) Retrieve the total quantity of books sold by each author:
152
153  SELECT b.author, SUM(o.quantity) AS Total_Books_Sold
154  FROM orders o
155  JOIN books b ON o.book_id=b.book_id
156  GROUP BY b.Author;
157
158
```

Data Output | Messages | Notifications

Showing rows: 1 to 314 Page No: 1 of 1

| | author character varying (100) | total_books_sold bigint |
|---|---|---|
| 1 | Jared Cortez | 10 |
| 2 | Tracy Parker | 11 |
| 3 | Taylor Wang | 9 |
| 4 | Cathy Knight | 6 |
| 5 | Bianca Matthews | 3 |
| 6 | Douglas Malone | 6 |
| 7 | James Alvarado | 9 |
| 8 | Betty Cross | 6 |

Total rows: 314 Query complete 00:00:00.151 CRLF Ln 156, Col 19

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**ADVANCED QUERY 7**

**Task to do:** List the cities where customers who spent over $30 are located

Query | Query History

```
153 SELECT b.author, SUM(o.quantity) AS Total_Books_Sold
154 FROM orders o
155 JOIN books b ON o.book_id=b.book_id
156 GROUP BY b.Author;
157
158
159
160 -- 7) List the cities where customers who spent over $30 are located:
161
162 SELECT DISTINCT c.city, total_amount
163 FROM orders o
164 JOIN customers c ON o.customer_id=c.customer_id
165 WHERE o.total_amount > 30;
166
167
```

Data Output | Messages | Notifications

Showing rows: 1 to 443 Page No: 1 of 1

| | city character varying (50) | total_amount numeric (10,2) |
|---|---|---|
| 1 | Taylorfurt | 189.45 |
| 2 | Leeport | 141.39 |
| 3 | Port Jasonview | 149.12 |
| 4 | Port Aaronstad | 145.44 |
| 5 | Matthewfurt | 328.50 |
| 6 | Angelaside | 42.19 |
| 7 | Lindaburgh | 325.92 |
| 8 | Stephanieberg | 156.60 |

Successfully run. Total query runtime: 133 msec. 443 rows affected.

Total rows: 443 | Query complete 00:00:00.133 | CRLF | Ln 165, Col 27

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No limit

ADVANCED QUERY 8

**Task to do:** Find the customer who spent the most on orders

Query Query History

```
161
162  SELECT DISTINCT c.city, total_amount
163  FROM orders o
164  JOIN customers c ON o.customer_id=c.customer_id
165  WHERE o.total_amount > 30;
166
167
168  -- 8) Find the customer who spent the most on orders:
169  SELECT c.customer_id, c.name, SUM(o.total_amount) AS Total_Spent
170  FROM orders o
171  JOIN customers c ON o.customer_id=c.customer_id
172  GROUP BY c.customer_id, c.name
173  ORDER BY Total_spent Desc LIMIT 1;
174
175
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

| | customer_id [PK] integer | name character varying (100) | total_spent numeric |
|---|---|---|---|
| 1 | 457 | Kim Turner | 1398.90 |

Total rows: 1 Query complete 00:00:00.113 CRLF Ln 173, Col 35

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ADVANCED QUERY 9

**Task to do:** Calculate the stock remaining after fulfilling all orders

Query | Query History

```
171 JOIN customers c ON o.customer_id=c.customer_id
172 GROUP BY c.customer_id, c.name
173 ORDER BY Total_spent Desc LIMIT 1;
174
175
176 --9) Calculate the stock remaining after fulfilling all orders:
177
178 SELECT b.book_id, b.title, b.stock, COALESCE(SUM(o.quantity),0) AS Order_quantity,
179     b.stock- COALESCE(SUM(o.quantity),0) AS Remaining_Quantity
180 FROM books b
181 LEFT JOIN orders o ON b.book_id=o.book_id
182 GROUP BY b.book_id ORDER BY b.book_id;
183
184
185
```

Data Output | Messages | Notifications

Showing rows: 1 to 500 | Page No: 1 of 1

| | book_id [PK] integer | title character varying (100) | stock integer | order_quantity bigint | remaining_quantity bigint |
|---|---|---|---|---|---|
| 1 | 1 | Configurable modular throughput | 100 | 3 | 97 |
| 2 | 2 | Persevering reciprocal knowledge ... | 19 | 0 | 19 |
| 3 | 3 | Streamlined coherent initiative | 27 | 5 | 22 |
| 4 | 4 | Customizable 24hour product | 8 | 0 | 8 |
| 5 | 5 | Adaptive 5thgeneration encoding | 16 | 8 | 8 |
| 6 | 6 | Advanced encompassing implem... | 2 | 0 | 2 |
| 7 | 7 | Open-architected exuding structure | 95 | 5 | 90 |
| 8 | 8 | Persistent local encoding | 84 | 3 | 81 |

Total rows: 500 | Query complete 00:00:00.266 | CRLF | Ln 182, Col 39

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